

Software Project Analysis: OCL & Sequence Diagrams

This report provides a detailed analysis of the Object Constraint Language (OCL) implementation and the operational flow of the system.

1. OCL Implementation: Preconditions, Postconditions, and Constraints

The system implements OCL concepts through a combination of Jakarta Validation and service-layer logic. Below are the formal OCL specifications for the `updateProfile` operation.

Operation: `updateProfile(userId: String, request: UpdateProfileRequest)`

Preconditions

Conditions that must be true before the operation is executed:

- **User Existence:** `pre: User.allInstances()->exists(u | u.id = userId)`
- **Valid Input:**
 - `pre: request.firstName.size() >= 2 and request.firstName.size() <= 50`
 - `pre: request.lastName.size() >= 2 and request.lastName.size() <= 50`
 - `pre: request.phoneNumber.matches('^\\+?[0-9]{7,15}$')`

Constraints (Invariants)

Conditions that must remain true throughout the operation:

- **Immutability of Identity:** `inv: self.id = self.id@pre` (The user's ID cannot change).
- **Immutability of Email:** `inv: self.email = self.email@pre` (The user's email cannot change).

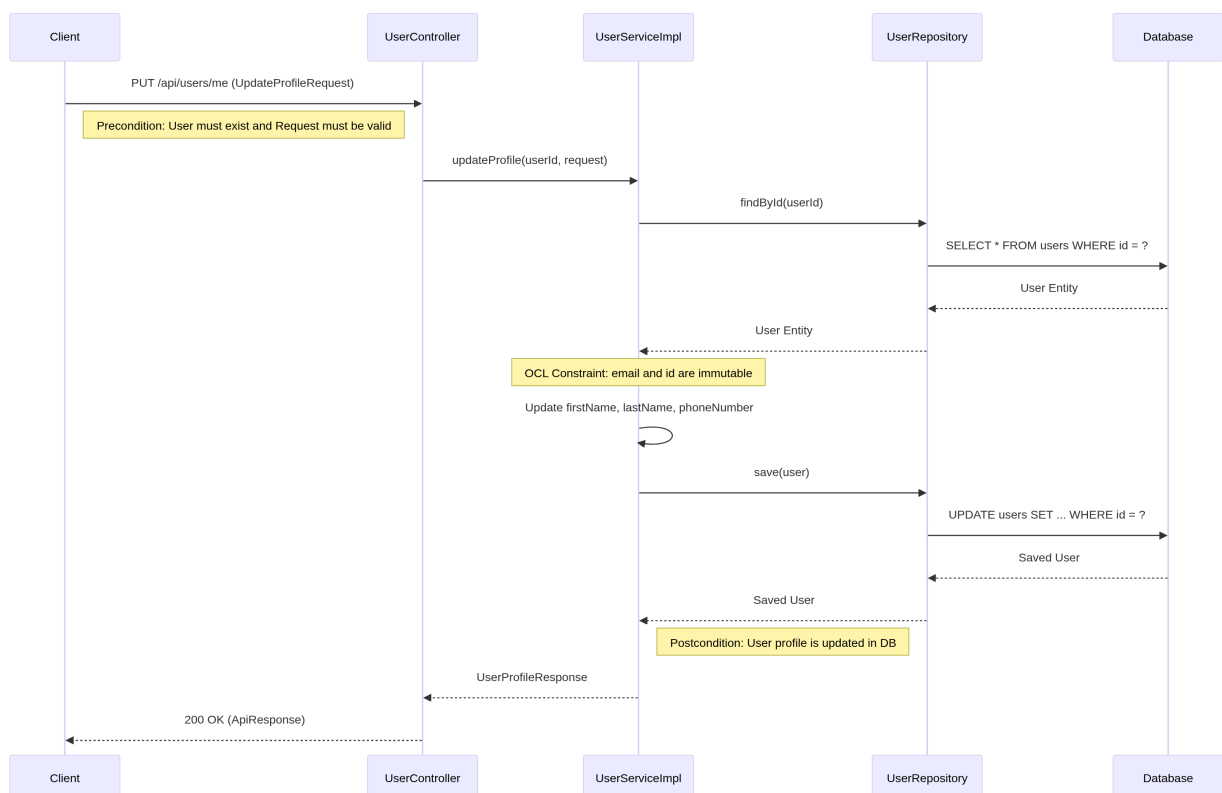
Postconditions

Conditions that must be true after the operation is completed:

- **Profile Updated:**
 - `post: self.firstName = request.firstName`
 - `post: self.lastName = request.lastName`
 - `post: self.phoneNumber = request.phoneNumber`
- **Audit:** `post: self.updatedAt >= self.updatedAt@pre`

2. Sequence Diagram: Update Profile Workflow

The following diagram illustrates the interaction between components during the profile update process, highlighting where OCL checks are performed.



For context, the structural relationships are maintained as follows:

